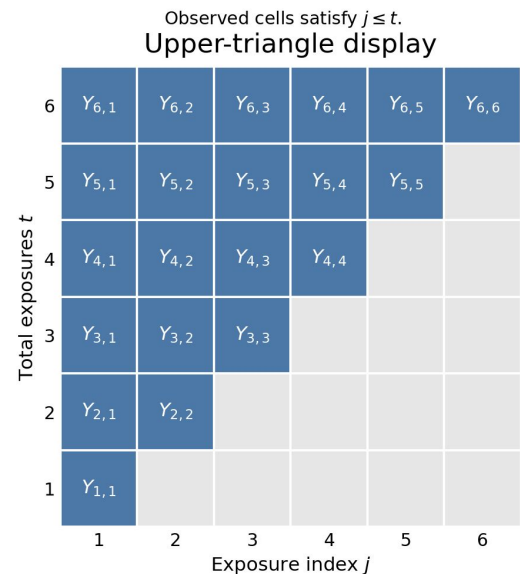


A/B Testing with Exposure-Dependent User Response

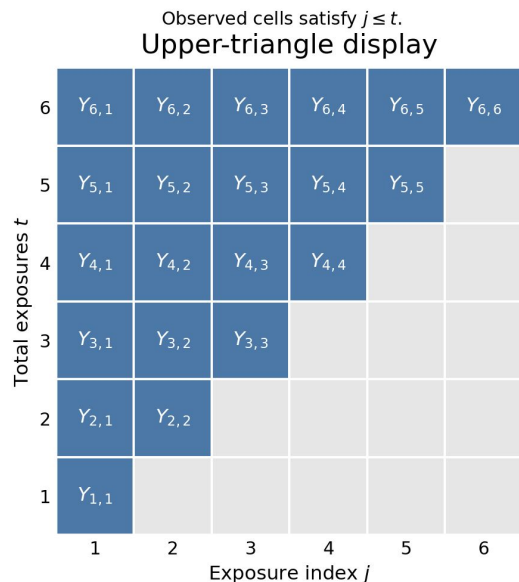
How does a **feature's effect** change with **repeated exposure**, and how long an exposure trajectory can we estimate reliably?

Scenario:

- introduce a new feature
- experiment window for 7 days
- observation: different users interacts with the feature a different number of times
- question: how does user interaction evolve over repeated interaction

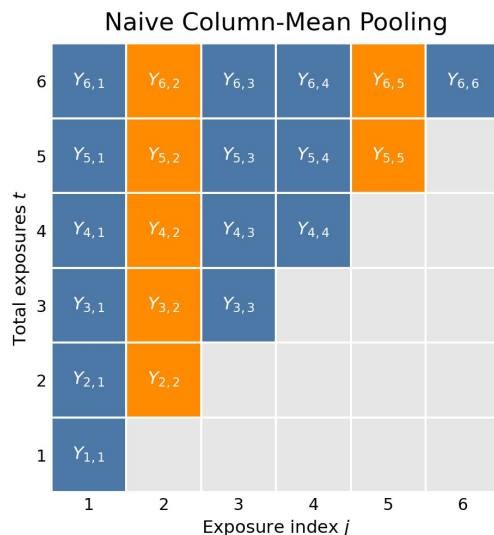


Individual "cells"



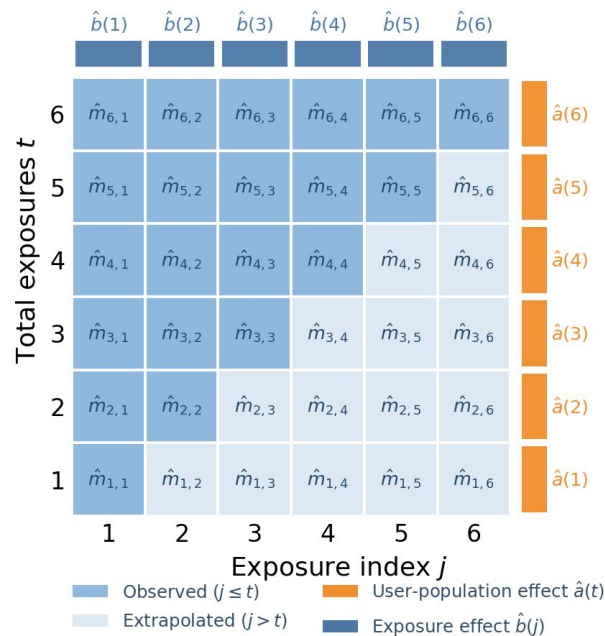
$O(B^2)$ parameters, where B denotes the maximum exposure count

Pooling by exposure



Potentially **biased** when user-response differs by total-exposure-count

Additive Fill: $\hat{m}_{t,j} = \hat{a}(t) + \hat{b}(j)$ for all cells

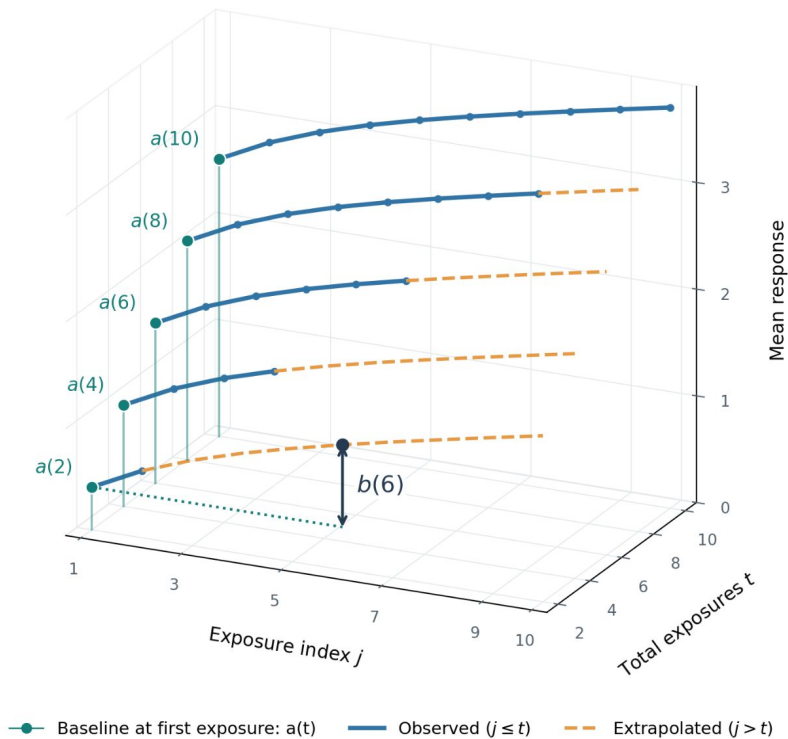


Parallel curve assumption to improve statistical efficiency
Allow different total-exposure groups to have different baseline

Different baselines, common exposure trend

$$m_{t,j} = a(t) + b(j), \quad b(1) = 0$$

$b(j)$: change from the first exposure, shared across groups

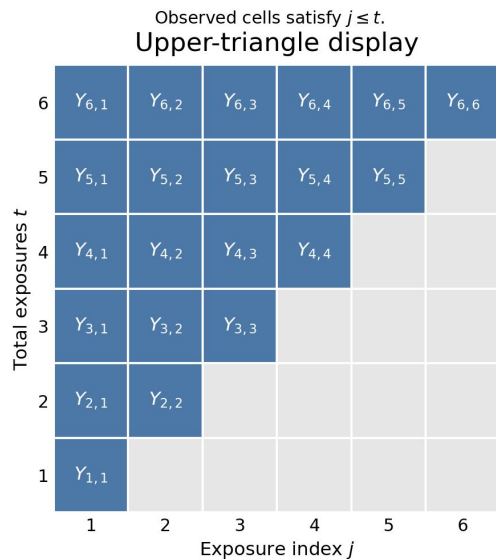


Illustrative model; baseline separation increased for clarity.

Challenge: parameters scales linearly in *maximum exposure count* (denoted by **B**)

Assume:

- 1) *monotonicity* of the exposure curve
- 2) *exposure-curve* consists of a *fixed number of jumps* (denote by **k**) that does not scale with B



Challenge: parameters scales quadratically in *maximum exposure count* (denoted by **B**)

Assume:

- 1) *monotonicity* of the exposure curve
- 2) *exposure-curve* consists of a *fixed number of jumps* (denote by k) that does not scale with B

Method:

$$\hat{\beta}_M = \Pi_{\mathcal{C}}^H(\hat{\beta}_U) = \operatorname{argmin}_{\beta \in \mathcal{C}} \|\beta - \hat{\beta}_U\|_H^2$$

$$\mathcal{C} : \text{monotone cone,} \quad \|v\|_H^2 = v^\top H v.$$

Mean-squared-error upperbound:

$$R_M = \frac{1}{B} \mathbb{E} [\|\hat{\beta}_M - \beta_0\|^2 \mid \mathcal{F}_T] \leq \sigma^2 k \log\left(\frac{eB}{k}\right) \left(\sum_{t=2}^B \frac{n_t}{N_t}\right) \left[\frac{1}{BN_B} + \sum_{j=2}^B \frac{1}{N_j j(j-1)} \right]$$

Under the assumption of $N_t \asymp n t^{-\alpha}$,

Exposure decay		Monotone risk upper bound
$0 < \alpha < 1$	poly-log in B	$R_M \lesssim \sigma^2 \log^2 B/n$
$\alpha = 1$	poly-log in B	$R_M \lesssim \sigma^2 \log^3 B/n$
$\alpha > 1$		$R_M \lesssim \sigma^2 B^{\alpha-1} \log^2 B/n$

Challenge: parameters scales quadratically in *maximum exposure count* (denoted by **B**)

Assume:

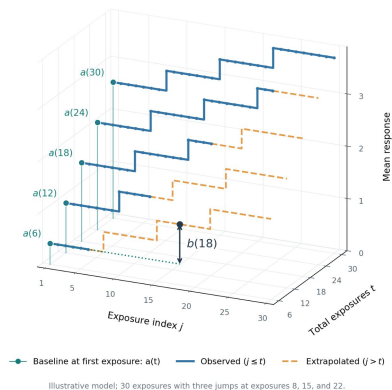
1) **monotonicity** of the exposure curve

2) *exposure-curve* consists of a **fixed number of jumps** (denote by k) that does not scale with

Different baselines, common exposure trend

$$m_{t,j} = a(t) + b(j), \quad b(1) = 0$$

$b(j)$: change from the first exposure, shared across groups



The "jump assumption" may not be a realistic modeling of the response curve

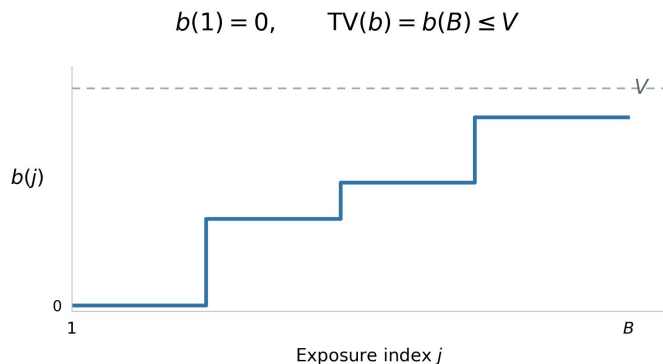
Challenge: parameters scales quadratically in *maximum exposure count* (denoted by **B**)

Assume:

1) **monotonicity** of the exposure curve

~~2) **exposure curve** consists of a **fixed number of jumps** (denote by k) that does not scale with~~

2) **total variation** of the exposure curve is bounded by V (i.e., $b(B) - b(1) \leq V$)



Mean-squared-error upperbound: for every $1 \leq k \leq B$,

$$\mathbb{E}[\|\hat{\beta}_M - \beta_0\|_H^2 \mid \mathcal{F}_T] \leq \underbrace{\frac{W_B V^2}{n k^2}}_{\text{approximation error}} + \underbrace{\frac{\sigma^2 \Lambda}{n} k \log \frac{eB}{k}}_{\text{estimation error}}.$$

Under the assumption of $N_t \asymp n t^{-\alpha}$,

Exposure decay

Monotone risk upper bound

$0 < \alpha < 1$

poly-log in B

$$R_M \lesssim \left(\frac{\sigma^2 V}{n}\right)^{2/3} B^{(1-\alpha)/3} \ell_B^{4/3} + \frac{\sigma^2 \ell_B^2}{n}$$

$\alpha = 1$

poly-log in B

$$R_M \lesssim \left(\frac{\sigma^2 V}{n}\right)^{2/3} \ell_B^{8/3} + \frac{\sigma^2 \ell_B^3}{n}$$

$\alpha > 1$

$$R_M \lesssim B^{\alpha-1} \left[\left(\frac{\sigma^2 V}{n}\right)^{2/3} \ell_B^{4/3} + \frac{\sigma^2 \ell_B^2}{n} \right]$$

Summary

- Proposed a **simple model** for user-response over exposure/interaction that aim to strike a balance between statistical efficiency and model bias
- Proposed the **projection estimator** on the monotone cone
- Established a finite-sample **MSE upperbound** over the projection estimator
- Provided a sufficient condition for **logarithmic dependency** of MSE on maximum-exposure-bound B